Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Backend Software Engineer and AI builder with 8+ years shipping production systems at scale - 1M+ property records, 100K+ monthly active users - and builder of **AgentProbe**: an open-source LLM observability and evaluation toolkit (LangChain, LangGraph, ChromaDB, FastAPI) that mirrors what LangSmith provides. Expert in Python, PostgreSQL, and AWS; advanced in multi-agent orchestration, RAG pipelines, and LLM evaluation frameworks. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

LLM Evaluation & Observability | LangChain / LangGraph | Multi-Agent Orchestration | RAG Pipelines / ChromaDB | Python (Backend) | FastAPI / REST APIs | PostgreSQL / SQL Tuning | AWS (S3, EC2, RDS) | Docker / CI/CD | High-Throughput Systems | Production Reliability | Agentic AI Systems

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Reduced deployment times by **40%** by building 20+ full-stack modules (Python, PostgreSQL) across 120+ Agile sprints - supporting **100K+ monthly active users** and **1M+ property records** with consistent sub-10 bug count for 6+ consecutive months.
- Cut microservices execution time by **50%** by leading Docker-based containerization for 30%+ of modules, delivering a scalable, observable backend architecture across multi-tenant production environments.
- Improved production query performance by **30%** by architecting PostgreSQL schemas, data access layers, and indexing strategies across 20+ projects - eliminating report-generation bottlenecks under high concurrent load.
- Achieved **100% PHI compliance** by migrating all client data to AWS S3 with automated pipelines, Libsodium encryption, and secure coding standards - zero security incidents; eliminated **80% of manual workload** via Python and PL-SQL automation.
- Improved system stability by **35%** and reduced critical issue backlog by resolving 100+ high-priority bugs, maintaining 99.5% uptime across a high-volume, multi-tenant platform.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered targeted mentoring in Python, data structures, algorithms, and database design for 30+ undergraduates - accelerating concept reinforcement through sprint-style feedback cycles.

PROJECTS

AgentProbe: LLM Observability & Evaluation Toolkit OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · Ollama · React · Azure

Built a **15-endpoint FastAPI evaluation backend** for LLMs, RAG systems, and multi-agent pipelines: 3 LangGraph agents (Orchestrator, RAG Tracer, Auto Fix), a smart LLM Router for dynamic tool routing, and a **5-metric eval engine** (factual accuracy, hallucination rate, retrieval quality, completeness, compliance). Supports Ollama, OpenAI, Anthropic, and Groq - runs 100% locally at zero cost. Integration test suite + React dashboard included. *github.com/venkatalakshmiparimala/AgentProbe*

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

Built a Flask + React app mapping natural language queries (EN/ES) to FHIR APIs using LLMs, embeddings, and BERT pipelines. HIPAA-compliant RBAC, schema-driven development, real-time dashboards. **Reduced manual retrieval time ~60%**; fully containerized, PHI-compliant. *github.com/venkatalakshmiparimala/AI_on_FHIR*

CrashClarity: UK Road Accident Analysis

Python · R · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Applied predictive modeling to 100K+ accident records; **85%+ accuracy** with Random Forest and Logistic Regression. Deployed interactive dashboards on GitHub Pages for non-technical stakeholders.

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & Agents: LangChain, LangGraph, LLM Router, ChromaDB, RAG, Embeddings, Ollama, OpenAI, Anthropic, Groq, FastAPI, Agentic AI, Prompt Engineering

Databases: PostgreSQL, MySQL, Oracle, NoSQL, GraphQL - schema design, query tuning, indexing, data access layers

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), RabbitMQ, Microservices

ML & Data: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, BERT, spaCy, Hugging Face, Feature Engineering, Predictive Modeling

Testing & Tools: TDD/BDD, Integration Testing, Selenium, Git, JIRA, Postman, Agile/Scrum, Figma